GitHub: https://github.com/qjmre23/Route2Zero | Live dashboard: https://route2zero.netlify.app/

Team Larpers Pilot Plan

Route2Zero Six-Month City Pilot

City pilot question

Can Route2Zero produce a city-owned, evidence-backed e-jeepney Phase-1 portfolio whose climate, equity, infrastructure, operator, and uncertainty assumptions have been tested against real local evidence?

PILOT PURPOSE

Convert Route2Zero 2.1 from an evidence-based prototype into a locally validated planning workflow that cities can inspect, challenge, refresh, and own.

PILOT SCOPE

- Target up to three partner LGUs, subject to city pairing and pilot agreement.
- Validate a manageable corridor sample, one flagship corridor, and a broader Phase-1 shortlist.
- Engage relevant operators/cooperatives, a utility or energy counterpart, and rider/community representatives where appropriate.
- Test ML service estimates, climate assumptions, route geometry, equity evidence, charging evidence, operator readiness, policy scenarios, robustness, and portfolio constraints.

SIX-MONTH OUTCOME

Outcome area	Deliverable
Decision	City-owned Phase-1 portfolio or shortlist with named next actions.
Evidence	Validated route cards, source registry, validation ledgers, and unresolved evidence owners.
Models	Calibrated service-intensity model, climate scenarios, model card, and evidence-confidence rules.
Operations	Scenario library, decision pack, refresh/retraining runbook, and reproducibility handover.

PILOT PATHWAY

PROTOTYPE	VALIDATE	CALIBRATE	CO-DESIGN	HANDOVER

Six months, three evidence gates

Timeline	Activity / Milestone	Core activity	Outcome
Month 1	Baseline + model protocol	Confirm decision owners and route sample; freeze source registry/build; approve validation, ML evaluation, privacy, and climate protocols.	Pilot charter; baseline build; route sample; validation protocol; risk register.
Month 2	Data refresh + ground truth	Extend the hackathon's 20-route OSM validation seed with field observation; validate geometry and service; collect operator, charging/depot, and local equity/accessibility evidence.	Expanded validation, operator, and charging ledgers; ground-truth sample; source refresh.
Gate 1	Evidence readiness review	Accept evidence plan and model-validation protocol; confirm minimum data coverage; resolve governance blockers.	Approved evidence gate or documented corrective actions.
Month 3	Model + route validation	Compare ML estimates with refreshed observations; recalculate metrics; review anomalies; tune geometry/evidence rules and equity variables.	Pilot model evaluation; model-card update; revised evidence grades; validated flagship case.
Month 4	Climate, charging + operator pre-feasibility	Calibrate diesel/electric assumptions; validate energy scenarios; request utility evidence; assess sites, operators, and verified constraints.	Calibrated climate scenario; charging/operator cards; optimizer-ready constraints.
Gate 2	Quality + route-set review	Accept pilot route set and model/evidence quality; hold routes with unresolved critical issues or mark them evidence-limited.	Approved route set with explicit holds and evidence limits.
Month 5	Scenario co-design + human factors	Run policy workshops; review sensitivity; agree portfolio constraints; test Priority vs Evidence vs Stability; review assistant and accessibility.	Approved scenarios; draft portfolio; usability and governance findings.
Month 6	Decision pack + handover	Finalize portfolio, evidence queue, model card, source registry, refresh process, issue log, analyst training, and task ownership.	Accepted decision pack; final build; reproducibility package; city handover.
Gate 3	Acceptance	City decision pack and reproducibility handover accepted by named decision owners.	Go-forward decision and accountable next-step owners.

Baseline traceability: build r2z-d4c8d4cc709a | scenario scn-e0f12f397e | service model service-v1-266e0b1d

The pilot succeeds when the evidence and decision improve

SUCCESS METRICS

Measure	Pilot success evidence
Coverage	% pilot routes ground-checked; % shortlisted routes with complete evidence cards; critical gaps resolved.
Model	MAE/RMSE vs refreshed observations, improvement over baseline, and documented route-level residual review.
Decision quality	Evidence-grade improvement; ranking/portfolio stability before vs after validation; routes moved from evidence-limited to decision-ready.
Adoption	LGU/operator/utility interactions; planner task completion; understanding of Priority, Evidence, and Stability.
Handover	One accepted city decision pack and one completed reproducibility/refresh handover.

VALIDATION METHODS

Field observations; operator interviews; LGU review; utility evidence requests; data reconciliation; grouped holdout testing; scenario workshops; usability/accessibility testing; stakeholder review of the flagship corridor.

RESPONSIBLE AI CONTROLS

- No resident-level personal data are required for core analytics; stakeholder data are consent-based.
- No hidden LLM score changes and no automatic funding or procurement decisions.
- No individual poverty or informal-status inference from population density.
- Stakeholders can challenge or replace evidence; an AI-disabled deterministic fallback remains available.

RISK REGISTER

Risk	Control / response
Historic route data	Use current-validation ledger and field checks; retain historic-only status until verified.
Weak ML performance	Prefer observed evidence; mark model experimental; compare with baseline.
Utility capacity unavailable	Show mapped proximity only; keep capacity NOT VERIFIED; request evidence.
Operator data unavailable	Keep neutral prior explicit and raise validation priority; a named desk reference is not readiness evidence.
Cost/financing data unverified	Treat feasibility figures as PROXY/SCENARIO until fleet and tariff data are confirmed; never present them as a budget.
Sparse geometry / equity data	Use reliability/evidence grades; validate traces; avoid unsupported settlement claims.
AI/API unavailable	Serve structured deterministic fallback; scores, scenarios, and portfolio are unaffected.
Constraint conflict	Show infeasibility and binding constraints; never fabricate a portfolio.

Validate the economics with the same discipline as the climate case

Starting-point scenario — not a budget

The Phase-1 shortlist currently implies about 1,943 vehicles, 102 chargers, and PHP 4.91 billion in vehicle-plus-charger hardware under explicit PROXY assumptions. Financing terms remain MISSING, and excluded costs are not silently absorbed.

FINANCIAL FEASIBILITY APPROACH

Input	Current status	Pilot validation needed
Fleet size	1,943 vehicles | PROXY	Confirm active fleet, duty cycle, service plan, spare ratio, and replacement phasing by corridor.
Vehicle cost	PHP 2.5M/unit | PROXY	Obtain supplier-neutral market evidence and define vehicle specification; do not treat the DOE planning value as a jeepney quote.
Charging	102 stations | PROXY	Validate charger power, depot layout, operating windows, utility capacity, interconnection, and civil works.
Financing	MISSING	Confirm tariff structure, equity/debt source, grants or subsidy, repayment terms, insurance, and ownership model.

Excluded from the current capital proxy: land or depot acquisition, civil works, distribution-system upgrades, interconnection fees, battery replacement, taxes, insurance, financing costs, and operations and maintenance.

DATA LICENSING & ATTRIBUTION

- Project code is MIT-licensed; source data retain their own licenses and conditions.
- OpenStreetMap route and infrastructure data require attribution to OpenStreetMap contributors and are used under ODbL. Attribution must remain visible in maps, exports, decision packs, and future field-data updates.
- The source registry records URLs, retrieval dates, license notes, checksums, and claim scope. Pilot owners review these before every release.
- New local evidence is accepted only with an accountable source, observation date, permission basis where relevant, and a field-level claim status.

SUBMISSION HEAD START

The evidence set includes 20 dated OSM route matches, 20 observed member-way geometries, and 22 usable source geometries. These records reduce the initial desk-review burden but do not prove active operations or franchise authority.

Baseline traceability: build r2z-d4c8d4cc709a | scenario scn-e0f12f397e | portfolio prt-fd6de9d793

GitHub: https://github.com/qjmre23/Route2Zero | Live dashboard: https://route2zero.netlify.app/

Named owners from data collection to city handover

TEAM STRUCTURE

Name	Role	Relevant background / experience / skills for this role
John Marwin Ebona	Product & Pilot Lead	City coordination, decision question, gates, integration, decision pack, handover.
Isaac Marcus	Engineering & Deployment Lead	Dashboard, scenarios, Netlify, Mapbox, reliability, and deterministic fallbacks.
Andrei Dela Cruz	Policy & Responsible AI Lead	Policy controls, decision rights, workshops, risk controls, and responsible AI.
Russel Mendez	Climate & Evidence Lead	Climate and energy method, source review, evidence confidence, and documentation.
Joaquin Sarmiento	Data & ML Lead	Source registry, feature store, service model, metrics, manifest, and model refresh.
Prince Marl Mirasol	QA & Documentation Lead	Testing, accessibility, artifact QA, claim checks, and issue tracking.
Carl Nueva	Geospatial & Infrastructure Analyst	Geometry review, spatial joins, equity layers, power evidence, and map QA.
JOSEPH CLARENCE PARAYAOAN	Field Validation & Partnerships Lead	Field plans, route checks, operator and partner coordination, and evidence handoff.

PARTNER RESPONSIBILITIES

Partner	Pilot responsibility
LGU	Define decision context, validate evidence, join scenario workshops, review outputs, accept handover.
Operator / cooperative	Validate operations, fleet/depot constraints, readiness, and feasibility.
Utility / energy	Review charging assumptions and flag formal capacity/interconnection studies.
Riders / community	Validate access concerns and challenge equity assumptions where relevant.

HANDOVER AND SCALE

Handover: final build, model card, source registry, field-observation template, validation ledgers, scenario library, decision pack, issue register, scheduled source-freshness check, refresh/retraining runbook, governance notes, and a city owner for each unresolved issue. Scale requires local adapters, recalibration, and validation; the Metro Manila model is not copied unchanged.